Aquila

Pressure Transient Analysis — Technical Methods

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This document specifies the analytical and numerical methods used by the Aquila pressure-transient analysis workbench. Every formula in the engine is referenced to its source, its dimensional form is given in oilfield units (the engine's canonical internal representation), and its validation status is stated. The document is intended for review by petroleum-engineering subject-matter experts and as a reference companion to the source code.

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1 Notation and units

The Aquila engine works internally in *oilfield* units throughout. Unit conversion happens only at the user-interface boundary (`modules/units.py`). This keeps every formula in the engine in a single consistent unit system and removes any risk of constant mis-translation across the codebase.

Symbol	Units	Meaning
k	mD	permeability
h	ft	net pay (formation thickness)
ϕ	—	porosity (fraction)
μ	cp	viscosity
B	rb/STB or rb/Mscf	formation volume factor
c_t	psi^{-1}	total compressibility
q	STB/D or Mscf/D	surface flow rate
ΔP	psi	pressure drawdown $p_i - p_{\text{wf}}$
Δt	hr	elapsed time
r_w	ft	wellbore radius
r_e	ft	drainage radius
s	—	mechanical skin (dimensionless)
C	bbl/psi	wellbore-storage coefficient
p_i	psia	initial reservoir pressure
T_R	°R	reservoir temperature (gas)
γ_g	—	gas specific gravity (air = 1)
Z	—	gas Z-factor (real-gas deviation)
$m(p)$	psi^2/cp	gas pseudo-pressure
t_a	psi·hr/cp	gas pseudo-time

2 The radial diffusivity equation

Aquila is built on the radial slightly-compressible-fluid diffusivity equation in a single homogeneous layer:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial p}{\partial r}\right) = \frac{\phi\mu c_t}{k}\,\frac{\partial p}{\partial t}. \tag{1}$$

The *line-source* solution to (1) with constant-rate inner-boundary condition and an infinite-acting outer boundary gives, at the wellbore,

$$\Delta P(t) \;=\; \frac{141.2\,q\,B\,\mu}{kh}\left[\tfrac{1}{2}\,\mathrm{E}_1\!\left(\frac{x}{1}\right) + s\right], \qquad x = \frac{948\,\phi\,\mu\,c_t\,r_w^2}{k\,t}, \tag{2}$$

where $\mathrm{E}_1(x) \equiv -\mathrm{Ei}(-x) = \int_x^\infty e^{-u}/u\;\mathrm{d}u$ is the exponential integral. In the late-time (small-x) limit, $\mathrm{E}_1(x) \to -\ln x - \gamma_{\mathrm{E}}$ where $\gamma_{\mathrm{E}} \approx 0.5772$ is the Euler–Mascheroni constant. This produces the famous *semilog approximation*:

$$\Delta P(t) \approx \frac{162.6\,q\,B\,\mu}{kh}\left[\log_{10} t + \log_{10}\!\left(\frac{k}{\phi\mu c_t r_w^2}\right) - 3.23 + \tfrac{1}{1.151}\,s\right]. \tag{3}$$

The slope of ΔP versus $\log_{10} t$ on this plot, $m \equiv 162.6\,qB\mu/(kh)$, is the foundation of conventional infinite-acting radial-flow (IARF) interpretation.

Bourdet derivative. The Bourdet (1989) derivative is

$$\frac{\partial \Delta P}{\partial \ln t} \;=\; t\,\frac{\mathrm{d}\Delta P}{\mathrm{d}t}. \tag{4}$$

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Under IARF, (3) gives $t\,\mathrm{d}\Delta P/\mathrm{d}t = m/\ln 10$, so the derivative is a *flat plateau* on a log-log plot. Departures from this plateau diagnose non-IARF flow regimes: wellbore storage, linear flow, pseudo-steady-state (closed-boundary), double-porosity transitions, layered behaviour, and so on. Aquila computes the derivative with a vectorised three-point log-time-window scheme:

$$\left(t\frac{\mathrm{d}\Delta P}{\mathrm{d}t}\right)_i = \frac{\Delta P_R \ln\frac{t_i}{t_L} - \Delta P_L \ln\frac{t_R}{t_i}}{\ln\frac{t_R}{t_i}\cdot\ln\frac{t_i}{t_L}}\cdot\frac{1}{\ln\frac{t_R}{t_L}}, \tag{5}$$

where t_L, t_R are the times within a log-window of half-width L to the left and right of t_i (default $L = 0.1$). The implementation is fully vectorised via `np.searchsorted` and is bit-identical to a naive loop but $\sim$1000$\times$ faster on a 6000-point trace.

3 Wellbore storage — Fair–Schulz model

Wellbore storage (WBS) dominates the early-time response. Pure WBS gives $\Delta P_{\mathrm{WBS}} = (qB/24C)\,t$ — a unit-slope ramp on log-log. Aquila uses the **Fair–Schulz** smooth transition between pure WBS and the line-source solution:

$$\Delta P(t) = \underbrace{w(t)}_{\text{weight}}\,\Delta P_{\mathrm{res}}(t) + \big[1 - w(t)\big]\,\Delta P_{\mathrm{WBS}}(t), \qquad w(t) = 1 - \exp\!\big[-(t/\tau)^2\big], \tag{6}$$

with characteristic time $\tau = 200\,C/(kh)$. The transition smoothly matches the unit-slope WBS region at early times to the line-source formation response at late times, and is differentiable so the Bourdet derivative remains smooth across the join.

The rule-of-thumb *clearance time* — after which the formation response dominates — is

$$t_{\mathrm{clear}} \;\approx\; (200\,000 + 12\,000\,s)\,\frac{C}{kh/\mu} \qquad [\mathrm{hr}]. \tag{7}$$

Aquila displays a warning when the planned test duration is less than $1.5\,t_{\mathrm{clear}}$.

4 Regression for IARF — solving for k and s

In a clean IARF window, the geometric mean of the Bourdet derivative yields the plateau, from which k follows directly:

$$\mathrm{plateau} = \exp\!\Big[\overline{\ln(t\,\mathrm{d}\Delta P/\mathrm{d}t)}\Big], \tag{8}$$

$$m = \ln(10)\cdot\mathrm{plateau} \qquad [\mathrm{psi/cycle}], \tag{9}$$

$$k = \frac{162.6\,q\,B\,\mu}{m\,h}. \tag{10}$$

Skin is recovered using the value of ΔP at $\Delta t = 1$ hr in the buildup. The standard Horner formula is

$$s \;=\; 1.151\left[\frac{\Delta P_{1\mathrm{hr}}}{m} \;-\; \log_{10}\!\left(\frac{k}{\phi\mu c_t r_w^2}\right) \;+\; 3.23\right]. \tag{11}$$

Quality flag — plateau flatness. The geometric standard deviation of $t\,\mathrm{d}\Delta P/\mathrm{d}t$ within the chosen regression window is reported:

$$f \;=\; \exp[\sigma(\ln(t\,\mathrm{d}\Delta P/\mathrm{d}t))] \tag{12}$$

Values $f < 1.15$ indicate a clean plateau; $f < 1.40$ is borderline; $f \geq 1.40$ indicates the window crosses a regime transition and the fit is suspect. The UI surfaces this directly to discourage the user from picking a window that includes wellbore storage or a boundary.

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5 Linear flow (fractured wells)

For a vertical well intersecting an infinite-conductivity fracture, the early-time response is *linear flow* into the fracture, with

$$\Delta P(t) = m_L \sqrt{t}, \qquad x_f \sqrt{k} \;=\; \frac{4.064\, q\, B}{m_L\, h} \sqrt{\frac{\mu}{\phi c_t}}, \tag{13}$$

where x_f is the fracture half-length. On the Bourdet derivative, linear flow appears as a *half-slope* line on log-log.

For a gas well, the equivalent pseudo-pressure formulation is

$$\Delta m(p)(t) = m_L \sqrt{t}, \qquad x_f \sqrt{k} \;=\; \frac{40.92\, q_g\, T_R}{m_L\, h} \sqrt{\frac{1}{\phi\, \mu_{\text{ref}}\, c_{t,\text{ref}}}}. \tag{14}$$

6 Closed-boundary (pseudo-steady-state, PSS)

After a well has drained out to a no-flow boundary, the pressure decline is uniform across the drainage area — *pseudo-steady-state*. On Cartesian axes, ΔP becomes a straight line in t:

$$\frac{\mathrm{d}\Delta P}{\mathrm{d}t} = m^* = \frac{0.234\, q\, B}{\phi c_t h\, A}, \qquad A = \pi r_e^2. \tag{15}$$

On a log-log Bourdet derivative, PSS appears as a *unit-slope* rising line at late time. Aquila's `regress_pss` extracts m^* by linear regression in real time and reports the drainage area, equivalent drainage radius $r_e = \sqrt{A/\pi}$, and pore volume:

$$V_p = A\, h\, \phi \;[\text{ft}^3] \quad \Longleftrightarrow \quad V_p\,[\text{bbl}] = V_p\,[\text{ft}^3]/5.615. \tag{16}$$

For gas wells, the slope and area formulae use the pseudo-pressure form with constant 2.356×10^6 in place of 0.234.

7 Gas-well analysis

For gas, the diffusivity equation is non-linear because μ, Z, and c_g all depend on pressure. The standard remedy is the Al-Hussainy / Ramey transformation to *pseudo-pressure*:

$$m(p) \;=\; 2 \int_0^p \frac{p'\, \mathrm{d}p'}{\mu(p')\, Z(p')}. \tag{17}$$

In the variable $m(p)$, the gas diffusivity equation is linear and has exactly the same form as (1). Hence every analytical solution carries over with the substitution $\Delta P \to \Delta m(p)$ and a different slope constant:

$$\Delta m(p)(t) = \frac{1422\, q_g\, T_R}{kh} \left[\tfrac{1}{2} \mathrm{E}_1(x) + s\right], \qquad m_g \equiv \frac{1637\, q_g\, T_R}{kh}. \tag{18}$$

For drawdown and buildup tests, time is also transformed:

$$t_a \;=\; \frac{t}{\mu(\bar{p})\, c_t(\bar{p})} \tag{19}$$

evaluated at the average reservoir pressure $\bar{p}$ (Agarwal approximation, accurate for buildup analysis when the pressure swing is small).

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7.1 PVT correlations

Aquila implements four standard correlations covering the practical $0.55 < \gamma_g < 1.7$, $1.0 < T_{pr} < 3.0$, $0.2 < P_{pr} < 30$ range:

Pseudo-criticals — Sutton (1985).

$$P_{pc} = 756.8 - 131.0\,\gamma_g - 3.6\,\gamma_g^2, \tag{20}$$

$$T_{pc} = 169.2 + 349.5\,\gamma_g - 74.0\,\gamma_g^2. \tag{21}$$

Acid-gas correction — Wichert–Aziz (1972). For mole fractions y_{CO_2} and y_{H_2S}:

$$A = y_{H_2S} + y_{CO_2}, \quad B = y_{H_2S}, \tag{22}$$

$$\varepsilon = 120\,(A^{0.9} - A^{1.6}) + 15\,(B^{0.5} - B^4), \tag{23}$$

$$T'_{pc} = T_{pc} - \varepsilon, \tag{24}$$

$$P'_{pc} = \frac{P_{pc}\,T'_{pc}}{T_{pc} + B(1-B)\varepsilon}. \tag{25}$$

Z-factor — Dranchuk–Abou-Kassem (1975).

$$Z \;=\; 1 + c_1\rho_r + c_2\rho_r^2 - c_3\rho_r^5 + c_4\big(1 + A_{11}\rho_r^2\big)\frac{\rho_r^2}{T_{pr}^3}\exp(-A_{11}\rho_r^2), \tag{26}$$

with $\rho_r = 0.27\,P_{pr}/(Z\,T_{pr})$ and DAK coefficients $A_1 \ldots A_{11}$ as in the 1975 paper. The eleven A_i are [0.3265, −1.0700, −0.5339, 0.01569, −0.05165, 0.5475, −0.7361, 0.1844, 0.1056, 0.6134, 0.7210]. The implicit relation in ρ_r is solved by bracketed bisection.

Viscosity — Lee–Gonzalez–Eakin (1966).

$$M_g = 28.96\,\gamma_g \qquad \text{[lbm/lbmol]}, \tag{27}$$

$$K = \frac{(9.4 + 0.02M_g)\,T_R^{1.5}}{209 + 19M_g + T_R}, \tag{28}$$

$$X = 3.5 + \frac{986}{T_R} + 0.01M_g, \qquad Y = 2.4 - 0.2X, \tag{29}$$

$$\mu_g = 10^{-4}\,K\,\exp\!\big(X\,\rho_g^{\,Y}\big) \qquad \text{[cp]}, \tag{30}$$

with the gas density ρ_g in g/cm^3 converted from $\rho_g[\text{lbm/ft}^3] = pM_g/(Z\,R\,T_R)$, $R = 10.7316$ psi·ft^3/(lbmol·°R).

Compressibility and FVF.

$$c_g \;=\; \frac{1}{p} - \frac{1}{Z}\left(\frac{\partial Z}{\partial p}\right)_T, \qquad B_g \;=\; 5.0287\,\frac{Z\,T_R}{p} \qquad \text{[rb/Mscf]}. \tag{31}$$

The $\partial Z/\partial p$ derivative is computed numerically with a 10^{-4} relative step.

7.2 Validation status

Correlation	Test point	Result
Sutton	$\gamma_g = 0.65$	$P_{pc} = 670$, $T_{pc} = 365$°R (matches paper)
Wichert–Aziz	$\gamma_g = 0.7$, 5%CO_2+10%H_2S	$\Delta T_{pc} = 20.7$°R
DAK	Standing–Katz, 8 points	7/8 within ±0.05
LGE	$p = 2000$ psia, $T = 200$°F	$\mu_g = 0.0168$ cp (in 0.014–0.020 band)
$m(p)$	monotonicity test	strictly increasing ✓
Forward → inverse k	three skin cases	k recovered to ≤ 0.05 %

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8 Multi-layer reservoirs

8.1 Commingled layers (no crossflow)

Two (or more) layers produce into the same wellbore but do *not* communicate vertically. Under the constraint of equal wellbore pressure and fixed total surface rate, the per-layer pressure responses combine by *parallel admittance*:

$$\Delta P_{\text{well}}(t) = \frac{q_{\text{total}}}{\sum_i Y_i(t)}, \qquad Y_i(t) = \frac{k_i h_i}{141.2\, B\, \mu \cdot g_i(t)}, \tag{32}$$

where $g_i(t) = \frac{1}{2}\mathrm{E}_1(x_i(t)) + s_i$ is the dimensionless line-source contribution of layer i evaluated at time t.

Identifiable parameters. From a single buildup, one can identify kh_{total} and the *flow-capacity ratio* $\kappa = k_1 h_1 / kh_{\text{total}}$, but the (k_i, h_i) pair separately is non-unique without independent rate-allocation data (e.g. production logging tools). Aquila surfaces this limitation explicitly in the inverse-fit output.

Signature on the Bourdet derivative.

- Early time: the fastest-responding layer dominates the derivative plateau (the high-k, low-storage layer).
- Late time: the derivative drops to the plateau corresponding to the system total kh.
- Transition: a downward *step* in the derivative — the qualitative fingerprint of commingled flow without crossflow.

8.2 Double permeability with crossflow

Pending verification against published reference type-curves. The numerical Laplace-inversion infrastructure (Stehfest 1970, validated to 0.30 % on the $K_0(\sqrt{s})/s$ radial-diffusion benchmark) is in place under `modules/laplace.py`. What remains is the Laplace-space transfer function $f(s; \kappa, \omega, \lambda)$ from Bourdet (1985) SPE 13628, which will be added once an authoritative reference is available for cross-validation.

9 Numerical Laplace inversion (Stehfest)

Several PTA solutions (notably double-permeability with crossflow, double-porosity, and bounded reservoirs with non-trivial outer geometries) are most naturally expressed in Laplace space. Aquila implements the Stehfest (1970) algorithm:

$$f(t) \;\approx\; \frac{\ln 2}{t} \sum_{i=1}^{N} V_i\, \bar{F}\!\left(\frac{i \ln 2}{t}\right), \tag{33}$$

with weights

$$V_i = (-1)^{N/2+i} \sum_{k=\lfloor (i+1)/2 \rfloor}^{\min(i,N/2)} \frac{k^{N/2}\,(2k)!}{(N/2-k)!\,k!\,(k-1)!\,(i-k)!\,(2k-i)!}. \tag{34}$$

Aquila uses $N = 8$ by default — sufficient for the smooth monotone transforms typical of radial PTA, while avoiding the catastrophic cancellation that affects $N = 14+$ on exponentially-decaying functions in double precision.

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Table 1: Stehfest inversion accuracy on benchmark transforms (max relative error over the indicated t-range).

Transform $\bar{F}(s)$	Inverse $f(t)$	Max error
$1/s^2$	t	1.5×10^{-4}
$\sqrt{\pi}/(2s^{3/2})$	$\sqrt{t}$	1.7×10^{-5}
$1/(s+a)$	e^{-at} (N=12)	2.0×10^{-3}
$\omega/(s^2+\omega^2)$	$\sin(\omega t)$ (N=12)	5.0×10^{-2}
$K_0(\sqrt{s})/s$	$\frac{1}{2}\mathrm{E}_1\left(\frac{1}{4t}\right)$	3.0×10^{-3}

Validation. Stehfest is validated against five known Laplace pairs (Table 1).

10 Uncertainty quantification

Aquila propagates input uncertainty on h and μ (oil) or h alone (gas — viscosity is PVT-derived) to the permeability estimate via Monte-Carlo sampling of triangular distributions. For each iteration i:

$$k_i = \begin{cases} \dfrac{162.6\, q\, B\, \mu_i}{m\, h_i} & \text{oil/water,} \\ \dfrac{1637\, q\, T_R}{m_g\, h_i} & \text{gas.} \end{cases} \tag{35}$$

The sample yields the petroleum-convention percentiles **P90** / **P50** / **P10** (low/best/high), a kernel-density-estimated PDF, the empirical CDF, and a Spearman-rank-correlation tornado sensitivity chart ranking the relative impact of each uncertain input.

11 Engine architecture

Listing 1: Aquila module layout.

```
aquila/
          app.py                    # Streamlit UI (orchestrator only)
          styles.py                 # Theming, CSS, Plotly layout
          assets/
                aquila_logo.svg       # Full logo (400x120)
                aquila_icon.svg       # Square favicon (100x100)
          modules/                  # Math layer     UI-independent
              engine.py             # Line-source, Fair-Schulz, Bourdet,
                                    #   regression (radial/linear/PSS).
              pvt.py                # DAK Z, LGE mu, Sutton+Wichert-Aziz,
                                    #   m(p) and t_a.
              layered.py            # Commingled (no crossflow);
                                    #   double-perm stub.
              laplace.py            # Stehfest numerical inverter.
              economics.py          # Cost & deferred-revenue calculator.
              uncertainty.py        # Monte Carlo + KDE/CDF + tornado.
              units.py              # Field <-> SI exact conversions.
              reporter.py           # FPDF report builder.
```

The math layer never imports Streamlit. Every module is independently unit-testable and reusable from a notebook or batch pipeline.

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Pressure transient analysis

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